DS	**Session**	Tree 2	**Question Information**	Level 1 Challenge 73

Question Description:

There are n cities and m flight connections between them. Your task is to add new flights so that it will be possible to travel from any city to any other city. What is the minimum number of new flights required?

Input

The first input line has two integers n and m: the number of cities and flights. The cities are numbered 1,2,…,n.

After this, there are m lines describing the flights. Each line has two integers a and b: there is a flight from city a to city b. All flights are one-way flights.

Output

First print an integer k: the required number of new flights. After this, print k lines describing the new flights. You can print any valid solution.

Constraints

- 1≤n≤10^5
- 1≤m≤2·10^5
- 1≤a,b≤n

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4 5
1 2
2 3
3 1
1 4
3 4
```

EXPECTED OUTPUT

```
1
4 1
```

Test Case 2

INPUT (STDIN)

```
4 5
1 3
2 1
1 2
2 4
1 4
```

EXPECTED OUTPUT

```
2
4 3
3 1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(m--)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
1
```

Test Case 2

TOKEN COUNT

```
1080
```

Test Case 3

NLOC

```
130
```

Code

C flight_expansion.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define MAXN 100005

int parent[MAXN];

int find_set(int v) {
    if (v == parent[v]) return v;
    return parent[v] = find_set(parent[v]);
}

void union_sets(int a, int b) {
    a = find_set(a);
    b = find_set(b);
    if (a != b) parent[b] = a;
}

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

    for (int i = 1; i <= n; i++) parent[i] = i;

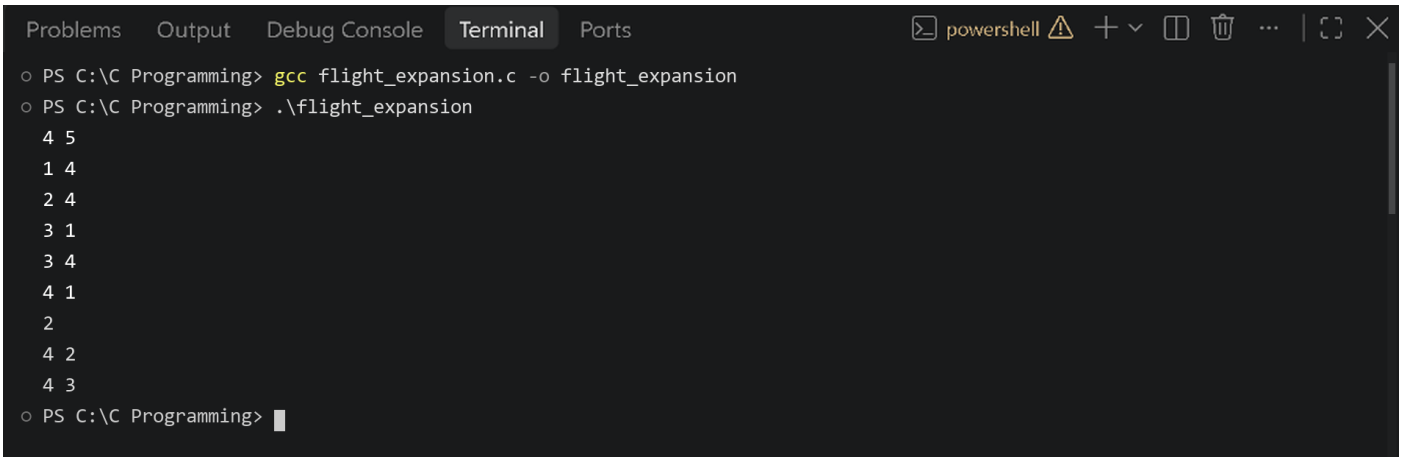
    int edges_left = m;
    while(m--) {
        int u, v;
        scanf("%d %d", &u, &v);
        union_sets(u, v);
    }

    int reps[MAXN];
    int comp_count = 0;
    for (int i = 1; i <= n; i++) {
        if (find_set(i) == i) {
            reps[comp_count++] = i;
        }
    }

    int k = comp_count - 1;
    printf("%d\n", k);
    for (int i = 0; i < k; i++) {
        printf("%d %d\n", reps[i], reps[i + 1]);
    }

    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc flight_expansion.c -o flight_expansion
PS C:\C Programming> .\flight_expansion
4 5
1 4
2 4
3 1
3 4
4 1
2
4 2
4 3
PS C:\C Programming>